Implementation of Subnetting in CISCO packet Tracer simulator

Exp. no: 9

Date: 22/8/26

Aim:

To implement subnetting in CISCO packet Tracer simulator

Introduction:

classless IP subnetting is a technique that allows for more efficient use of IP addresses by allowing for subnet masks that are not just the default masks for each IP class.

Steps for Creating a subnetting

(i) The first step is to create a network topology.

(ii) To create a topology in packet Tracer, select the 'New' button in the top left corner, there select 'Network' and 'Generic'

(iii) Once, after Creating a topology, we can add devices.

(iv) Add devices by dragging them from the bottom left corner into the topology.

(v) Then Connect them by dragging cable from one devices port to another devices port.

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Subnetting

To subnet the network address of 192.168.1.0/24 to provide enough space for atleast 5 addresses for end devices, the switch and the router, we can use a /27 subnet mask. This will give us 8 subnets with 30 host address each.

The IP addressing for the network shown in the topology can be as follows:

- Router R1:
- GigabitEthernet0/0: 192.168.1.1
- GigabitEthernet0/1: 192.168.2.1
- Switch S1:
- FastEthernet0/1: 192.168.1.0/27
- PC1: 192.168.1.11
- PC2: 192.168.1.12
- PC3: 192.168.1.13
- PC4: 192.168.1.14
- PC5: 192.168.1.15
- FastEthernet0/2: 192.168.2.0/27
- PC1: 192.168.2.11
- PC2: 192.168.2.12
- PC3: 192.168.2.13
- PC4: 192.168.2.14
- PC5: 192.168.2.15
- Router R2:
- FastEthernet0/0: 192.168.3.1
- FastEthernet0/1: 192.168.4.1
- Switch S2:
- FastEthernet0/1: 192.168.3.0/27
- PC1: 192.168.3.11
- PC2: 192.168.3.12
- PC3: 192.168.3.13
- PC4: 192.168.3.14
- PC5: 192.168.3.15
- FastEthernet0/2: 192.168.4.0/27
- PC1: 192.168.4.11
- PC2: 192.168.4.12
- PC3: 192.168.4.13
- PC4: 192.168.4.14
- PC5: 192.168.4.15

Configuring the Devices :-

(1) Configuring Router

* Right click on the router and select 'CLI'.
* In the CLI (Command line Interface), enter the following commands.

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enable
configure terminal
interface Fast Ethernet 0/0
ip address {Ip address} {subnet mask}
no shutdown
exit.

* The first interface, Fast Ethernet 0/0, will be connected to the switch, while the second interface, Fast Ethernet 0/1. will be connected to one of the pc's.

(ii) Configuring switch

* Next configure the switch, by right clicking the switch and select CLI. Enter the following commands:

enable
configure terminal
interface Fast Ethernet 0/1
switch port mode access
exit
interface Fast Ethernet 0/2
switch port mode access
exit.

(iii) Configuring pc's

Now configure the pc's. Right click on the pc's and select 'Config'. In the Configuration

window, enter IP address, subnet mask, default gateway and DNS server information.

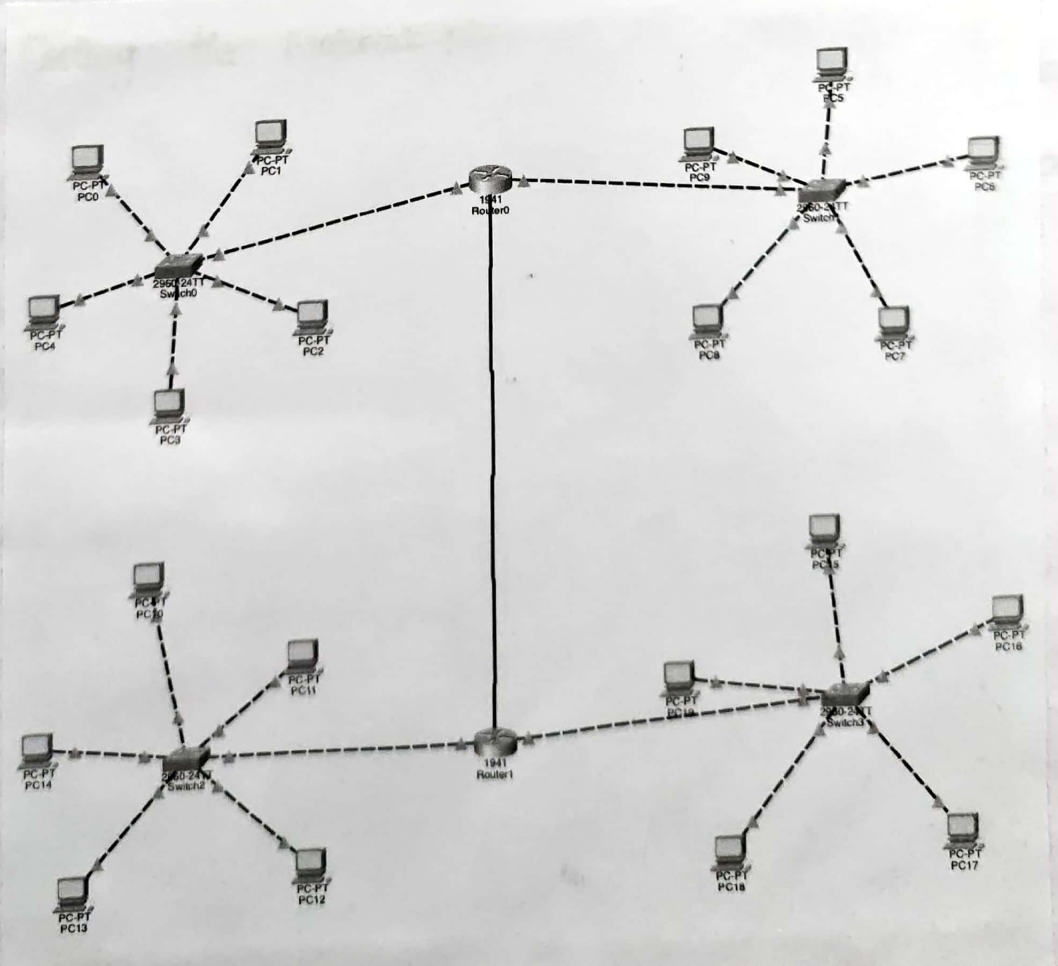
(iv) Configuring Gigabit Ethernet

To configure Gigabit ethernet interface on the router, follow these steps:

(i) Right click on the router, click CLI.

(ii) Enter the following commands.

```
# enable
# configure terminal
# interface Gigabit Ethernet 0/0
# ip address {IP address} {subnet mask}
# no shutdown
# exit.
```



Testing the network :

* Open a command prompt on each pc and try to ping other pc.
* If the ping is successfull, then the network is functioning properly.

Fire	Last Status	Source	Destination	Type	Color	Time(sec)	Periodic	Num
●	Successful	PC4(2)	Router2	ICMP		0.000	N	12
●	Successful	PC4(2)	PC2(1)(1)	ICMP		0.000	N	13
●	Successful	PC0	Router0	ICMP		0.000	N	14

Result :

Thus, the subnetting was implemented successfully in CISCO packet Tracer simulator.

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